Year 10 Science
Physical Sciences
Worksheet - Acceleration

ACCELERATION

Acceleration occurs whenever the velocity of a body changes.

Acceleration is defined as the rate of change of velocity.

For the purpose of examples considered in this text acceleration is considered to be uniform (or constant).

From the definition:

$$\text{Average acceleration} = \frac{\text{change in velocity}}{\text{time}}$$

$$= \frac{\text{final velocity} - \text{initial velocity}}{\text{time}}$$

$$a(av) = \frac{v - u}{t}$$

rearranging gives: $\boxed{v = u + at}$ (assuming uniform acceleration)

where: $v =$ final velocity
$u =$ initial velocity
$a =$ uniform acceleration
$t =$ time interval (measured from zero)

UNITS OF ACCELERATION

As acceleration is the rate of change of velocity then its units are those of velocity divided by time.

Units: $\frac{m\ s^{-1}}{s} = m\ s^{-2}$ (or m/s^2)

TYPE EXAMPLE 10: Determine the acceleration of a body which is accelerated from 2 $m\ s^{-1}$ to 10 $m\ s^{-1}$ in 2.5 s.

$u = 2\ m\ s^{-1}$
$v = 10\ m\ s^{-1}$
$a = ?$
$t = 2.5\ s$

$$v = u + at$$
$$\therefore a = \frac{v - u}{t} = \frac{(10 - 2)}{2.5} = \frac{8}{2.5} = 3.2\ m\ s^{-2}$$

i.e. acceleration is 3.2 $m\ s^{-2}$ (in the direction of the initial motion).

TYPE EXAMPLE 11: What is the initial velocity of a body which attained a velocity of 16 $m\ s^{-1}$ when accelerated at 1.5 $m\ s^{-2}$ for 8 s?

$u = ?$
$v = 16\ m\ s^{-1}$
$a = 1.5\ m\ s^{-2}$
$t = 8\ s$

$$v = u + at$$
$$\therefore u = v - at = 16 - (1.5 \times 8) = 16 - 12 = 4\ m\ s^{-1}$$

i.e. the initial velocity is 4 $m\ s^{-1}$ (in the direction of the final velocity)

SET 8: ACCELERATION

1. What is the increase in velocity in each second of a body accelerated at 1.8 $m\ s^{-2}$?
2. Calculate the final velocity of a motor car initially travelling at 10 $m\ s^{-1}$ which is accelerated at 1.4 $m\ s^{-2}$ for 10 s.
3. A body starts from rest and is accelerated at 2.7 $m\ s^{-2}$. What is the velocity after 12 s?
4. A ball rolling down a slope from rest is accelerated at 3.5 $m\ s^{-2}$. What is the ball's velocity after 6 s?
5. Find the acceleration of a plane which increases its velocity from 100 $m\ s^{-1}$ to 200 $m\ s^{-1}$ in 40 s.
6. What is the average acceleration of a body which is accelerated from rest to 15 $m\ s^{-1}$ in 5 s?
7. Determine the average acceleration experienced by a body which increases its velocity from 7 $m\ s^{-1}$ to 34 $m\ s^{-1}$ in 9 s.
8. A vehicle starting from rest is accelerated at 2.5 $m\ s^{-2}$. Determine the time taken to reach a velocity of 5 $m\ s^{-1}$.
9. Find the time taken for a body travelling with a velocity of 15 $m\ s^{-1}$ to reach a velocity of 29 $m\ s^{-1}$ when accelerated at 0.7 $m\ s^{-2}$.
10. A car is accelerated from 10 $m\ s^{-1}$ to 25 $m\ s^{-1}$ in 2.5 s. What is the average acceleration?

NEGATIVE ACCELERATION

If a body (travelling in the positive direction) slows down uniformly it can be seen from the equation derived from the definition of acceleration that the value obtained for acceleration will be negative.

That is, as $a = \frac{v - u}{t}$ then if $v < u$, a will be negative.

Whenever the velocity vector and the acceleration vector are in opposite directions the magnitude of the velocity decreases uniformly.

A negative acceleration is also referred to as *RETARDATION* or *DECELERATION.*

TYPE EXAMPLE 12: A car is uniformly slowed down from 26 $m\ s^{-1}$ to 6 $m\ s^{-1}$ in a period of 8 s. Calculate the acceleration of the car.

$u = 26\ m\ s^{-1}$
$v = 6\ m\ s^{-1}$
$t = 8\ s$
$a = ?$

$$\begin{aligned} v &= u + at \\ \therefore a &= \frac{v - u}{t} \\ &= \frac{6 - 26}{8} \\ &= \frac{-20}{8} \\ &= -2.5\ m\ s^{-2} \end{aligned}$$

i.e. acceleration is $-2.5\ m\ s^{-2}$ (or $2.5\ m\ s^{-2}$ in the opposite direction to the initial motion).

NOTE: When given the rate of slowing down or the retardation assign a negative value to the magnitude (e.g. a body decelerated at 2 $m\ s^{-2}$ has an acceleration of $-2\ m\ s^{-2}$).

SET 9: NEGATIVE ACCELERATION

1. A train approaching a railway station at 30 m s^{-1} is brought to rest in 10 s. Determine the retardation.
2. A plane on landing touches down at a speed of 80 m s^{-1}. Determine the acceleration if the plane comes to rest in a time of 8 s.
3. A truck travelling at 36 m s^{-1} is decelerated uniformly at 1.5 m s^{-2}. Calculate the: (a) velocity after 6 s.
   (b) time taken to come to rest.
4. A plane flying at 170 m s^{-1} slows down uniformly at 2 m s^{-2}. What is the plane's velocity after 30 s?
5. How long will it take a vehicle travelling at 30 m s^{-1} to reach a velocity of 12 m s^{-1} if the acceleration is -3 m s^{-2}?
6. A bus travelling at 25 m s^{-1} is decelerated at 2.5 m s^{-2}. Find the time taken for the bus to come to rest.
7. A motor cyclist uniformly decelerates at 3 m s^{-2}. If the initial velocity is 32 m s^{-1}, calculate the time taken to attain a velocity of 14 m s^{-1}.
8. A cyclist brakes suddenly with a uniform acceleration of -4 m s^{-2} and comes to rest in 2.5 s. Determine the initial velocity of the cyclist.

SET 8

1. 1.8 m s^{-1}	**2.** 24 m s^{-1}	**3.** 32.4 m s^{-1}	**4.** 21 m s^{-1}	**5.** 2.5 m s^{-2}
6. 3 m s^{-2}	**7.** 3 m s^{-2}	**8.** 2 s	**9.** 20 s	**10.** 6 m s^{-2}

SET 9

1. 3 m s^{-2}	**2.** -10 m s^{-2}	**3.** **(a)** 27 m s^{-1}	**(b)** 24 s	**4.** 110 m s^{-1}
5. 6 s	**6.** 10 s	**7.** 6 s	**8.** 10 m s^{-1}	